

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks: 20

Annexure A

Descriptive Written Test

Code of Conduct:

I T. Subahan (192511121) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: -Subahan

75
100

15
20

Q

Smart Home Automation System and Intelligent Agents:

A Smart home automation system controls devices such as lights, air conditioners, heaters and security cameras based on user behaviour and environmental conditions. The system receives inputs from sensors such as motion detectors, temperature sensors, and clocks. Intelligent agents help the system make decisions automatically.

Types of Intelligent Agents:

1. Simple Reflex Agent:

A simple reflex agent makes decisions based only on the current state of environment. It uses predefined rules and does not consider previous experiences.

Example:

- If motion is detected, turn on the lights.
- If the temperature exceeds 30°C , switch on the air conditioner.
- If no motion is detected, turn off lights.

Advantages:

1. Fast response
2. Easy to implement
3. Requires less memory

Disadvantages:

1. Cannot learn user preferences
2. Ignore past experience

advantages:

1. Flexible and adaptive
2. makes intelligent decisions
3. Suitable for complex environments

4. Utility-Based Agent:

A utility based agent selects actions that maximize overall satisfaction or utility.

example:

- User comfort
- Energy efficiency
- Security
- Cost

advantages:

1. Balances multiple objectives
2. Learns user preferences
3. Optimizes overall performance.

Disadvantages:

1. Designing utility functions is difficult.
2. Computationally expensive

2. Robot Navigation in an Unknown Maze

A robot is placed in an unknown maze and must find the exit without prior knowledge of the environment. Search algorithms help the robot explore possible paths and reach the goal efficiently.

Uniform Cost Search (UCS):

Uniform cost search expands the node with the lowest path cost first.

Working:

1. Start from the initial position
2. Calculate the cost of each possible path
3. Expand the path with min cost
4. Continue until the exit is reached.

example:

→ Path A: cost = 5

→ Path B: cost = 8

The robot selects path A because it has a lower cost.

Advantages:

- Finds the optimal path.
- Works with different path costs
- Guarantees the shortest solution.

Disadvantages:

- Requires large memory
- Slow for large mazes

Complexity:

- Time complexity: $O(b^d)$
- Space complexity: $O(b^d)$

Where:

b = branching factor
 d = depth of solution

Iterative Deeping Search (IDS):

Iterative Deeping Search combines depth first search and breadth first search.

Working:

1. Search to depth 1.
2. If the goal is not found, search to depth 2.
3. Continue increasing the depth limit.
4. Stop when the exit is found.

Advantages:

- Require less memory.
- Guarantees completeness

Disadvantages:

- Repeats node expansion
- more computation time.